

INTERVENTION-CONDITIONED LAYER FRAGILITY ACROSS LANGUAGE-MODEL PRETRAINING

Anonymous authors

Paper under double-blind review

ABSTRACT

Interventions on neural-network components are often interpreted as measuring properties such as redundancy or robustness, yet recent work shows that the conclusion can depend on the intervention protocol. We test whether such protocol dependence can be reduced to differences in output-level functional damage. Our assay measures the fraction of intact next-token top-1 predictions preserved after an interior-block intervention across independently pretrained small PYTHIA models. The early-to-late block-bypass response replicates across nine 160M runs, a held-out run, a 70M scale, and WikiText-2 and HellaSwag-derived evaluation streams. Within residual attenuation, matched comparisons show that KL/NLL predictive damage discriminates top-1 damage more strongly than raw activation displacement. However, prospective joint KL/NLL matching does not make block bypass and norm-controlled activation noise interchangeable: a corrected family residual of -0.0168 remains across five runs. A frozen block of simple output-logit and decision-boundary features accounts for only 20.2% of that residual. These results support a bounded methodological conclusion: in the tested setting, intervention responses are conditioned on how a model is perturbed and cannot be summarized by raw magnitude or matched predictive damage alone. We do not identify a causal mechanism, establish downstream capability loss, or claim generality beyond the tested small-model regime.

1 INTRODUCTION

Interventions—removing a layer, patching an activation, or injecting noise—are widely used to infer redundancy, necessity, robustness, or causal roles in neural networks. Such interpretations are strongest when reasonable intervention procedures agree and weaker when the response depends on the measurement protocol.

This concern is established empirically. Language models can preserve many next-token predictions after layer deletion or swapping (Lad et al., 2025), while Garcia (2026) shows that output-grounded layer equivalence changes across training and differs between replacement and interchange. Intervention audits also show dependence on the comparator and matching procedure (Luo et al., 2026). We therefore ask a narrower question: *can protocol-dependent responses be explained by differences in functional predictive damage?*

The explanation is plausible. Nominally equal interventions can change output distributions differently, while interventions far apart in activation space can be close in function space. If matching forward KL and target-token NLL removes protocol dependence, intervention identity adds little beyond a severity budget. If not, raw size and those output-level measures are insufficient summaries.

We study independently pretrained 70M- and 160M-parameter PYTHIA models (Biderman et al., 2023). For a fixed corpus and checkpoint, we intervene on an interior block and measure the retained fraction S of intact next-token top-1 predictions; $D_S = 1 - S$ is top-1 damage and $\Delta S = S_{\text{late}} - S_{\text{early}}$ is the training endpoint change. This operational metric is not new: it belongs to the relative-accuracy family (Lad et al., 2025) and complements the divergent-token rate on identical positions (Deiseroth et al., 2024).

The block-bypass endpoint is negative in all nine 160M runs, including a prospectively held-out run, and repeats at 70M and on a HellaSwag-derived stream. This establishes a stable outcome, not novelty

for the broad training phenomenon. Within residual attenuation, complementary matching shows that KL/NLL damage is more informative than raw activation displacement, without establishing KL/NLL causality.

The central test prospectively targets activation-noise strengths, using only KL and NLL, to match block interventions by run, checkpoint, and layer. After a tie-handling inconsistency was withdrawn and repaired, the canonical analysis retains a family residual of -0.0168 (run-bootstrap 95% interval $[-0.0200, -0.0126]$): matched activation noise produces less top-1 damage than block attenuation on average. A frozen output-logit and decision-boundary feature block reduces this residual by 20.2% but does not eliminate it.

Our contributions are exactly three:

1. We establish an independently replicated empirical base for a fixed block-bypass response across two small PYTHIA scales and two English evaluation streams, including held-out, confidence, and layer controls.
2. Complementary magnitude- and damage-matched contrasts show that output-level predictive damage is more informative than raw activation displacement within the frozen residual-attenuation assay.
3. Prospective joint KL/NLL targeting and a corrected deterministic top-1 analysis show that predictive-damage matching does not eliminate the block/noise difference; simple output geometry explains only part of it.

We do not infer universal specialization, large-model scaling, downstream capability loss, KL/NLL causality, or an identified internal mechanism.

2 RELATED WORK

Layer redundancy and protocol dependence. Layer deletion and swapping can preserve substantial next-token behavior (Lad et al., 2025); depth-pruning criteria likewise remove some final-model layers with limited aggregate degradation (Men et al., 2025). Our closest neighbor is Garcia (2026), who shows that replacement and interchange yield different output-grounded gaps across Pythia checkpoints. That work occupies the broad protocol- and training-dependence claim. We ask whether two intervention families become interchangeable after matching output KL and NLL across independent pretraining runs.

Training-dependent perturbation fragility. Quantization degradation varies across pretraining and learning-rate phases (Catalan-Tatjer et al., 2026); activation-noise fragility can change after probe accuracy saturates (Reblitz-Richardson, 2026), and layer-specific noise reveals heterogeneous learning dynamics (Borras et al., 2022). We therefore claim neither generic novelty for training-dependent fragility nor a smooth training law.

Parameter, activation, and function space. Network composition, normalization, location, and direction decouple raw perturbation size from functional change. Conversely, norm dominates recent gradient-free language-model adaptation experiments in their specific regime (Kim et al., 2026). Our result rejects only a pure raw-displacement account within the frozen attenuation assay. Forward KL, NLL change, and top-1 damage share logits, so their alignment remains descriptive without controlled contrasts.

Mechanistic and matched interventions. Activation patching and editing depend on corruption, metric, and intervention design (Zhang & Nanda, 2023; Makelov et al., 2024); self-repair adds further architecture- and prompt-specific dependence (Rushing & Nanda, 2024). SteerCheck is the closest methodological collision: it prospectively matches off-target KL across steering comparators and retains direction/family structure (Luo et al., 2026). We do not claim matched-KL auditing as new; we target joint KL/NLL for training-conditioned block substitutability and add independent-run replication and explicit assay repair.

Prediction-disagreement metrics. The divergent-token metrics of Deiseroth et al. (2024) measure argmax changes under pruning and quantization. On identical teacher-forced positions, our S is one minus such a disagreement rate; novelty is claimed only for the controlled result built around it.

3 EXPERIMENTAL SETUP

Models, data, and statistical units. We study independent PolyPythias replicas from the PYTHIA suite (Biderman et al., 2023): nine 160M runs form the principal trajectory, five 70M runs provide scale replication, and five or six 160M runs support the controlled comparisons. The independent pretraining run is the scientific replicate; tokens, layers, pairs, and evaluation selections are not treated as independent. The primary WikiText-2 and cross-corpus HellaSwag-derived streams are fixed teacher-forced sequences with the same tokenizer and 4,608 predicted positions per checkpoint. Exact run/checkpoint allocations and selection details are in Appendix A.

Interventions. For block output $b(x)$ with residual input x , residual attenuation is

$$\tilde{b}_\alpha(x) = b(x) - \alpha [b(x) - x], \quad (1)$$

with $\alpha \in \{0.25, 0.50, 0.75, 1.00\}$; $\alpha = 1$ is block bypass. We intervene on blocks 1–10 of the 12-block 160M architecture. The second family adds a frozen random direction after the intact block with scale β normalized by the local activation norm and averages directions within cell.

Agreement and training response. For intact distribution p and intervened distribution q_I on fixed teacher-forced positions t , define

$$S(I, c) = |T|^{-1} \sum_{t \in T} \mathbb{K}[\hat{y}_{c,t} = \hat{y}_{I,c,t}], \quad (2)$$

$$D_S(I, c) = 1 - S(I, c), \quad (3)$$

$$\Delta S(I) = S(I, c_{\text{late}}) - S(I, c_{\text{early}}), \quad (4)$$

where each $\hat{y}$ is the lowest-index token among exact maximum logits. The core statistic averages over frozen interior layers and evaluation selections; negative ΔS means lower retained top-1 agreement late in training.

Damage and magnitude. We measure

$$D_{\text{KL}}(I) = \mathbb{E}_t \text{KL}(p_t \| q_{I,t}), \quad (5)$$

$$D_{\text{NLL}}(I) = \text{NLL}(q_I) - \text{NLL}(p). \quad (6)$$

Raw magnitude is the local relative residual-change norm; absolute activation RMS is a sensitivity measure. These are descriptors, not assumed causes.

Matching and uncertainty. Within residual attenuation, magnitude matching constrains relative norm while requiring functional-damage separation; damage matching calipers joint KL/NLL while requiring location or magnitude separation. Pairing is deterministic, within run/checkpoint, and without reuse. For the cross-family test, each block cell defines a joint KL/NLL target. An outcome-blind deterministic search uses only noise strength, KL, NLL, and technical diagnostics; strict ‘Class A’ requires both damage targets to meet their frozen relative calipers. The primary residual is

$$R_{\text{family}} = \text{mean}_r \text{median}_{i \in r} [D_S^{\text{noise}}(i) - D_S^{\text{block}}(i)], \quad (7)$$

with practical-equivalence band ± 0.010742 . Exact calipers are reported in Appendix A. Intervals bootstrap independent-run summaries; fixed intact-confidence bins, exclusions, and contrasts were frozen before confirmatory outcomes.

4 A REPRODUCIBLE INTERVENTION-CONDITIONED RESPONSE

We first establish that the outcome used for mechanism discrimination is not a single-checkpoint or single-run accident. This section is an empirical base for the functional-damage question; it is not presented as discovery of the broad training phenomenon.

Table 1: Selected validated results. Intervals bootstrap independent-run summaries.

Result	Estimate	Replicates
WikiText-2 ΔS	$-0.1035 [-0.1115, -0.0958]$	9/9
HellaSwag ΔS	$-0.0793 [-0.0885, -0.0677]$	6/6
Magnitude-matched contrast	$+0.0430 [0.0354, 0.0525]$	6/6
Damage-matched contrast	$+0.0025 [-0.0009, 0.0058]$	4/6
Corrected family residual	$-0.0168 [-0.0200, -0.0126]$	5/5
Geometry-adjusted residual	$-0.0134 [-0.0176, -0.0092]$	5 runs

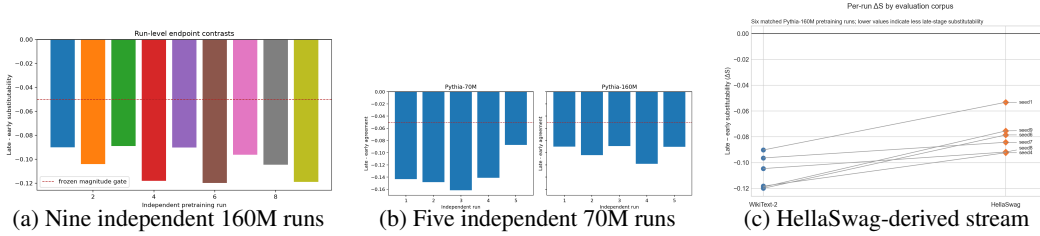


Figure 1: **Empirical base.** The endpoint direction repeats across independent runs, two small scales, and two evaluation streams. Panels reuse scaled ARC figures without recomputing outcomes. Differences in magnitude are not interpreted as invariance or a scaling law.

Independent runs and held-out prediction. Across nine independent 160M pretraining runs, every late-minus-early block-bypass change is negative. Mean ΔS is -0.10353 , with a run-bootstrap 95% interval $[-0.11149, -0.09585]$; the median is -0.10410 . Run 9 was prospectively held out before its weights were evaluated. Its $\Delta S = -0.11895$ has the predicted sign and lies inside the frozen prediction interval $[-0.13463, -0.06366]$. Intact NLL improves from an early-run mean of 3.878 to a late-run mean of 3.726, so the endpoint is not explained by a failure to learn next-token prediction.

The direction is broad across the tested locations: all 90 run–interior-layer endpoint changes are negative, and all ten layer medians are negative. The magnitude is not uniform; layer medians range from -0.0653 to -0.1838 . Accordingly, we claim broad sign support, not location invariance.

Confidence control. Top-1 agreement is mechanically related to intact confidence. We therefore repeat the endpoint comparison within five fixed intact-confidence bins. All 45 eligible run–bin changes are negative, with mean matched change -0.11388 . This control conditions on a coarse observable and does not fully match token difficulty.

Scale and corpus boundaries. Five independent 70M runs reproduce the direction (5/5 negative; median $\Delta S = -0.14334$, 95% interval $[-0.15385, -0.11030]$). This is replication at two small scales inside one architecture family, not a scaling law.

On the HellaSwag-derived stream, all six 160M runs are negative, with mean $\Delta S = -0.07926$ and 95% interval $[-0.08851, -0.06767]$. All 30 confidence- bin changes are also negative. The absolute effect is 0.766 times the WikiText-2 mean (95% interval $[0.645, 0.880]$): the direction repeats but the magnitude is corpus-dependent. HellaSwag is easier and more confident under the frozen stream, so the comparison does not imply corpus invariance.

Taken together, these checks justify using the endpoint response as a stable outcome for the controlled comparisons that follow. They do not establish why the response changes.

5 FUNCTIONAL DAMAGE EXPLAINS MORE THAN RAW MAGNITUDE

We next discriminate between two simple descriptions of the response. A raw- magnitude account predicts that interventions moving the local activation by a similar amount should produce similar top-1 damage. A functional-damage account instead predicts that the change in output distribution is the

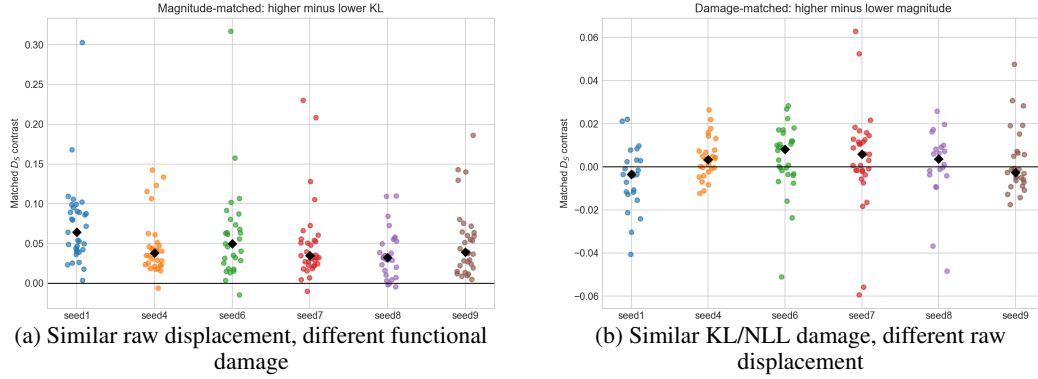


Figure 2: **Complementary matched contrasts within residual attenuation.** Functional-damage differences retain a stable top-1 contrast at matched magnitude, while the residual magnitude contrast is small after joint KL/NLL matching. Points and intervals summarize independent runs, not tokens.

more informative quantity. Because all variables are observational consequences of an intervention, we use matched contrasts rather than interpreting raw correlations.

5.1 MAGNITUDE-MATCHED COMPARISON

The primary arm contains 185 disjoint pairs formed within run and checkpoint. Their median relative-magnitude ratio is 1.019, while the median absolute KL and NLL-damage separations are 0.087 and 0.098. In 94.6% of pairs, the higher-KL member also has higher NLL damage. The mean run-median difference in top-1 damage is $+0.04297$ (95% interval $[0.03537, 0.05246]$), positive in all six runs and all three prospectively confirmatory runs. An absolute-RMS matching sensitivity is stronger rather than weaker.

Thus, approximately equal local displacement does not equalize the response when functional damage differs. The contrast holds within all five intact- confidence bins, although fixed bins are not exact propensity matches.

5.2 DAMAGE-MATCHED COMPARISON

The reverse arm contains 162 disjoint pairs with median absolute KL and NLL gaps 0.0050 and 0.0070. Their median magnitude ratio is 1.129. The mean run-median top-1 contrast is $+0.00246$, with a 95% interval $[-0.00090, 0.00575]$. A stricter subset of 37 pairs with magnitude ratio at least 1.25 has contrast -0.00054 and interval $[-0.00456, 0.00371]$. Checkpoint-level signs in this stricter subset are heterogeneous.

The two contrasts have a clear ordering: in this attenuation family, functional-damage separation survives raw-magnitude matching, whereas raw-magnitude separation leaves little stable contrast after joint KL/NLL matching. This weakens a pure raw-displacement account. It does not show that raw magnitude is generally irrelevant: the observational regression retains a magnitude coefficient, location retains secondary structure, and the second intervention family behaves differently.

Finally, KL, NLL, and D_S are computed from the same intervened logits. The matched design shows discrimination among measured descriptors; it does not randomize KL/NLL or establish mediation.

6 FUNCTIONAL DAMAGE DOES NOT ELIMINATE INTERVENTION-FAMILY DEPENDENCE

The within-family result motivates the paper’s central test. If output-level damage is an adequate description of an intervention, then two structurally different interventions matched on KL and NLL should have equivalent top-1 damage. We compare block attenuation with norm-controlled additive activation noise after the intact block.

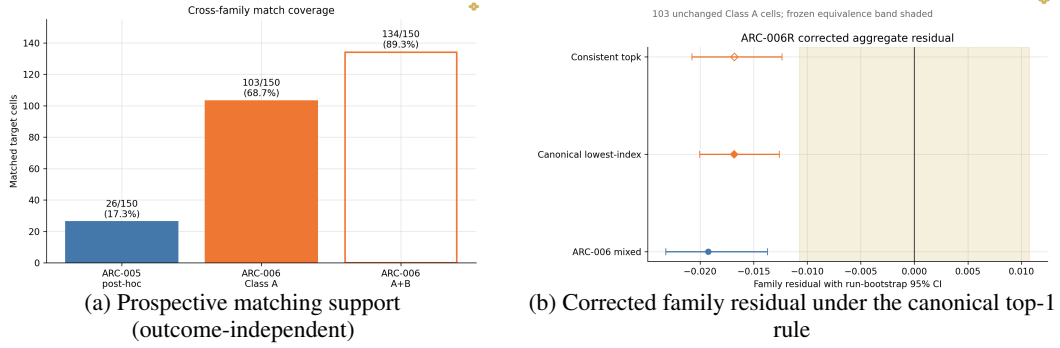


Figure 3: **Predictive-damage matching does not eliminate family dependence.** Panel (a) is a matching diagnostic and remains valid because the assay repair did not change assignments. Panel (b) uses only corrected ARC-006R outcomes; the withdrawn mixed-rule estimate is not shown as evidence.

Qualitative replication in a second family. The noise family reproduces the early-to-late direction: all five independent runs have negative ΔS , with mean -0.07246 and 95% interval $[-0.08438, -0.06304]$; all 25 eligible confidence-bin comparisons are negative. This rules out the narrow claim that the sign is unique to literal block bypass. It does not imply that the two families have the same quantitative response. An initial post-hoc cross-family comparison in ARC-005 had only 26/150 strict matches and failed its match-quality gate; its residual is not used as confirmatory evidence.

Prospective joint-damage targeting. For each block target at the same run, checkpoint, and layer, a blinded one-dimensional search chooses the noise strength using only KL, NLL, strength, and technical diagnostics. It never computes or records D_S . Strict Class A support rises to 103/150 cells (68.7%); the A+B sensitivity set contains 134/150 (89.3%). Mean relative KL and NLL errors among Class A targets are 1.94% and 1.87%, respectively. Matching assignments and calipers are frozen before top-1 outcomes are revealed.

Assay invalidation and repair. The first reveal was invalid. Historical block outcomes had used `'topk(2)[...,0]'`, whereas new noise outcomes used `'argmax'`. Exact FP16 ties can resolve to different token indices; the maximum single-cell correction was comparable with the invalid aggregate. Analysis stopped before downstream geometry modeling. The original estimate is withdrawn.

ARC-006R changes only the identity convention: both families select the lowest vocabulary index among exact maximum logits. It does not change targets, matching classes, seeds, checkpoints, layers, equivalence bounds, exclusions, or the statistical protocol. The repair shifts the invalid point estimate by $+0.00240$ (12.49% of its magnitude), demonstrating that the bug was scientifically material.

Corrected residual. Under the canonical rule,

$$R_{\text{family}} = -0.016833, \quad (8)$$

$$95\% \text{ CI} = [-0.020045, -0.012616]. \quad (9)$$

All five run medians are negative and every leave-one-run-out estimate remains below the negative equivalence boundary. Because the residual is $D_S^{\text{noise}} - D_S^{\text{block}}$, its negative sign means that block attenuation changes more top-1 predictions than matched activation noise on average, despite similar KL/NLL damage. The interval lies wholly outside the frozen ± 0.010742 equivalence band. The A+B sensitivity estimate is -0.01816 and is directionally consistent.

The aggregate is not cellwise universal: 24/103 corrected cells reverse sign, and layer means range from negative to positive. The result is therefore a run-level family component with substantial heterogeneity, not a deterministic law for every layer and checkpoint.

The safe conclusion is limited but consequential: in this assay, KL and NLL alignment is not sufficient to make the two interventions behaviorally exchangeable. The result does not identify which unmeasured property carries the difference.

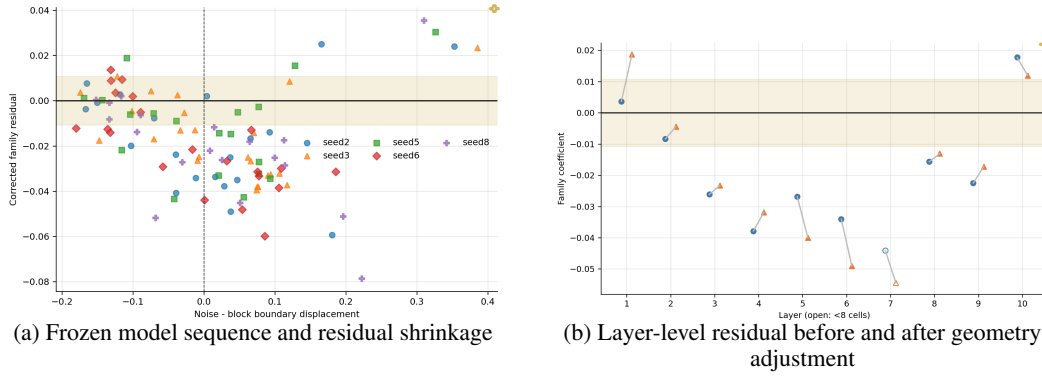


Figure 4: **Output geometry is associated with only part of the family residual.** The full frozen geometry block reduces the aggregate coefficient, but the adjusted residual and heterogeneous layer structure remain. These are associational controls, not a causal decomposition.

7 SIMPLE OUTPUT GEOMETRY IS NOT SUFFICIENT

The family residual may still be a simple artifact of the top-1 decision rule. KL and NLL summarize the full predictive distribution, whereas D_S depends on whether a perturbation crosses the local argmax boundary. We therefore test a frozen sequence of non-tautological output-geometry models on the corrected 103-cell support set.

The baseline model conditions on the functional-damage summary and fixed run/checkpoint/layer effects. Adding the intact top-1 margin alone does not change the family coefficient. Adding anchored boundary displacement also fails to shrink it: the estimated magnitude grows by 1.49%. A full block that adds logit-change norm and absolute alignment reduces the family coefficient by 20.18%, from approximately -0.0168 to -0.01344 (95% interval $[-0.01764, -0.00923]$). The point estimate remains outside the frozen equivalence region, although its interval overlaps the boundary slightly.

The partial shrinkage is accompanied by counterevidence to a simple boundary story. An outcome-blind geometry-matched subset covers 31/103 cells and retains a residual of -0.02436 (95% interval $[-0.03006, -0.01819]$). The geometry features do not pass the frozen sign-reversal prediction gate (leave-one-run-out AUC 0.664), and layer heterogeneity increases by 26.2% after adjustment rather than decreasing. The far-margin stratum is near zero, but the effect is largest at medium rather than nearest margins.

We conclude that the tested output geometry is incomplete. We do not conclude that a deeper internal direction causes the remainder.

8 DISCUSSION AND LIMITATIONS

8.1 INTERVENTIONS AS CONDITIONAL MEASUREMENTS

Raw displacement is insufficient within residual attenuation, and functional damage is more informative, yet joint KL/NLL matching leaves a stable block/noise difference. Intervention identity therefore retains information not captured by either severity description; it is not itself an explanation. A layer may look substitutable under one procedure and not another because the procedures trace different paths through activation space. Intervention claims should consequently report the model, data, family, magnitude convention, functional-damage budget, response metric, and support diagnostics rather than describe protocol-independent “layer redundancy.”

8.2 WHAT REMAINS UNIDENTIFIED

The two families induce strongly different hidden perturbation directions, but frozen observational direction features do not pass their preregistered explanatory gates. A planned counterfactual direction

test fails its support gate before top-1 outcome reveal. We therefore make no causal claim about internal direction; the blinded feasibility results remain in Appendix J.

8.3 LIMITATIONS

Scope. Evidence is limited to 70M/160M PYTHIA replicas, two English evaluation streams with one tokenizer, and two intervention families; cross-family damage matching is 160M and WikiText-2 only. Corpus magnitudes differ by about 23%, and replacement, interchange, attention-only, MLP-only, editing, other architectures, and billion-parameter models remain untested.

Construct validity and heterogeneity. S is discontinuous teacher-forced top-1 agreement, not semantic equivalence, downstream capability, or pruning safety. KL, NLL, geometry, and D_S share logits, so matched associations may retain common metric geometry. Although the aggregate family residual is seed-stable, 24/103 cells reverse sign and layer effects vary; low support at layer 7 limits location-specific interpretation.

Mechanism and assay repair. The frozen tests weaken raw-magnitude, family-invariant damage, and simple output-geometry accounts but do not identify an internal mechanism. The first prospective reveal was invalidated by inconsistent FP16 top-1 tie handling; we stopped, withdrew it, froze a family-independent lowest-index rule, and recomputed without changing matches or exclusions. The corrected conclusion survives, but the 12.49% point-estimate change is material; full provenance is in Appendix F.

9 CONCLUSION

We tested whether protocol-dependent layer-intervention responses reduce to raw magnitude or output-level predictive damage. Across independent small PYTHIA runs and two streams, functional KL/NLL damage is more informative than raw displacement within attenuation, yet prospective damage matching leaves a corrected block/noise residual that simple output geometry only partly explains. The mechanism remains unidentified; under the tested conditions, intervention responses are protocol-conditioned measurements rather than protocol-independent layer properties.

AI USE STATEMENT

Generative AI systems assisted with research-workflow orchestration, literature triage, code generation and review, experiment-execution support, analysis checks, and manuscript drafting and formatting. All cited literature, reported numerical results, and manuscript claims were checked against preserved sources and experiment artifacts; invalidated and inconclusive analyses were not used to support the paper’s claims. Generative AI was not used to fabricate data, citations, or unexecuted results. The authors reviewed all AI-assisted work and take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

ETHICS STATEMENT

This study analyzes publicly released language models and evaluation corpora. It does not involve human participants, deployment, or external communication. The principal research-integrity risk is overinterpreting intervention assays; we address it by preserving invalidated analyses, separating associative from causal evidence, and limiting claims to the tested models, corpora, and interventions.

REPRODUCIBILITY STATEMENT

Section 3 defines the frozen assay and statistical units. The appendix provides run-level summaries, matching diagnostics, confidence controls, cross-corpus details, the assay invalidation and deterministic top-1 repair, layer analyses, and artifact provenance. The accompanying research artifacts retain preregistrations, commands, raw and processed results, and figure-generating inputs for the validated analyses.

REFERENCES

- Stella Biderman et al. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. URL <https://proceedings.mlr.press/v202/biderman23a.html>.
- Hendrik Borras, Bernhard Klein, and Holger Froning. Walking noise: On layer-specific robustness of neural architectures against noisy computations and associated characteristic learning dynamics. *arXiv preprint arXiv:2212.10430*, 2022. URL <https://arxiv.org/abs/2212.10430>.
- Albert Catalan-Tatjer, Niccolo Ajroldi, and Jonas Geiping. Training dynamics impact post-training quantization robustness. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2510.06213>.
- Bjorn Deiseroth, Max Meuer, Nikolas Gritsch, Constantin Eichenberg, Patrick Schramowski, Matthias Assenmacher, and Kristian Kersting. Divergent token metrics: Measuring degradation to prune away LLM components—and optimize quantization. In *Proceedings of NAACL-HLT*, 2024. URL <https://aclanthology.org/2024.naacl-long.377/>.
- Gabriel Garcia. No free swap: Protocol-dependent layer redundancy in transformers. *arXiv preprint arXiv:2605.16234*, 2026. URL <https://arxiv.org/abs/2605.16234>.
- Taeyeon Kim, Ahhyun Kim, TaeHyeon Kim, and Unggi Lee. Not the dimension, the norm: What matters in gradient-free weight perturbation of language models. *arXiv preprint arXiv:2608.01624*, 2026. URL <https://arxiv.org/abs/2608.01624>.
- Vedang Lad, Jin Hwa Lee, Wes Gurnee, and Max Tegmark. The remarkable robustness of LLMs: Stages of inference? In *Advances in Neural Information Processing Systems*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/bcad07d4bfab51243efaa08b8ed475b3-Abstract-Conference.html.
- Daming Luo, Christy Liang, and Junyu Xuan. SteerCheck: Attribution specificity and alignment leakage in activation-steering audits. *arXiv preprint arXiv:2608.24335*, 2026. URL <https://arxiv.org/abs/2608.24335>.
- Aleksandar Makelov, George Lange, Atticus Geiger, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Ebt7Qd7Jwy>.
- Xin Men, Mingyu Xu, Qingyu Zhang, Qianhao Yuan, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. ShortGPT: Layers in large language models are more redundant than you expect. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. URL <https://aclanthology.org/2025.findings-acl.1035/>.
- Orion Reblitz-Richardson. When probing accuracy saturates, fragility resolves: A complementary metric for LLM pre-training analysis. *arXiv preprint arXiv:2606.11375*, 2026. URL <https://arxiv.org/abs/2606.11375>.
- Cody Rushing and Neel Nanda. Explorations of self-repair in language models. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/rushing24a.html>.
- Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023. URL <https://arxiv.org/abs/2309.16042>.

A FULL RUN TABLES

The machine-readable source of every number appearing in the main text is ‘paper/numerical-provenance.csv’. Full run-level outputs remain in the sealed ARC directories rather than being copied or rounded again. The principal 160M run table is ‘meta_arc05/ARC-20260826-5060-003/results/run_level_results.csv’; the cross-corpus table is ‘research/arc_20260828_5060_011/processed/core_run_results.csv’.

Frozen run and checkpoint allocation. The principal trajectory uses nine 160M runs at steps 14k, 36k, 72k, 107k, and 143k; scale confirmation uses five 70M runs. Within-family mechanism comparisons use six 160M runs (pilot labels 1, 4, and 9; confirmatory labels 6, 7, and 8) at steps 14k, 72k, and 143k. Cross-family targeting uses runs 2, 3, 5, 6, and 8 at those three checkpoints.

Frozen selection and matching details. Three selection seeds each produce six sequences with 256 predicted positions, giving 4,608 positions per checkpoint. Magnitude-matched pairs require relative magnitude ratio at most 1.10 and absolute KL separation at least 0.03. Damage-matched pairs require KL gap at most $\max(0.01, 0.10\bar{D}_{\text{KL}})$ and NLL gap at most $\max(0.015, 0.10\bar{D}_{\text{NLL}})$, plus layer separation at least three or magnitude ratio at least 1.25. Cross-family targeting uses a deterministic grid and bounded one-dimensional refinement; top-1 outcomes remain hidden until the target manifest and analysis are sealed.

B SEED-LEVEL ROBUSTNESS

The primary WikiText-2 endpoint is negative in 9/9 160M runs. All leave-one-run-out medians remain negative. The held-out ninth run is negative in all three evaluation selections. At 70M, 5/5 independent runs are negative. On the HellaSwag-derived stream, 6/6 runs and 18/18 fixed evaluation resamples are negative. For the corrected family residual, all five run medians are negative and all leave-one-run-out estimates remain below the negative equivalence boundary.

C MATCHING DIAGNOSTICS

Within-family matching is performed without pair reuse and within run and checkpoint. Primary magnitude matching yields 185 pairs; damage matching yields 162 pairs. Cross-family targeting obtains 103/150 strict Class A cells and 134/150 A+B cells. The strict-set mean relative KL and NLL target errors are 1.94% and 1.87%. Sixteen targets remain Class C. The worst target has 16.45% relative KL and 23.42% relative NLL error and is excluded by the frozen class rule, not by its outcome.

D CONFIDENCE CONTROL

Fixed intact-confidence bins are $[0, .05)$, $[.05, .1)$, $[.1, .2)$, $[.2, .4)$, $[.4, 1.01)$. The WikiText-2 core has 45/45 negative eligible run-bin endpoint changes. The activation-noise confirmation has 25/25 negative run-bin changes. The HellaSwag-derived stream has 30/30 negative changes. Fixed bins reduce a known composition difference but do not exactly match token difficulty.

E CROSS-CORPUS DETAILS

Both streams use 4,608 predicted positions per checkpoint and the same tokenizer and sequence-selection procedure. WikiText-2 contains encyclopedia prose; the HellaSwag-derived stream contains correct everyday-event continuations. Mean intact NLL is lower and intact confidence higher on the latter. Its mean absolute endpoint effect is 0.766 of the WikiText-2 reference, with a bootstrap interval $[0.645, 0.880]$. The functional-damage versus raw-magnitude ordering repeats, but the damage-matched raw-magnitude contrast is small and above zero on HellaSwag. We therefore say “replicates across the tested corpora,” not “corpus invariant.”

F ASSAY INVALIDATION AND DETERMINISTIC TOP-1 REPAIR

ARC-006 compared historical block outcomes extracted with ‘topk’ against new noise outcomes extracted with ‘argmax’. Exact FP16 ties can choose different indices. ARC-007 detected the discrepancy at its reproduction gate before any geometry model was fit. The original -0.019235 residual and its reversal counts are invalid.

ARC-006R freezes a family-independent rule: select the lowest token index among exact maximum logits. It reuses the same 150 targets, classes, seeds, checkpoints, layers, calipers, and equivalence

band. The corrected primary residual is -0.016833 ; a globally consistent ‘topk’ sensitivity is -0.016797 . Their difference is below the frozen numerical-instability threshold. The repair, source hashes, and output manifest are retained in the ARC-006R directory.

G LAYER HETEROGENEITY

The core endpoint sign is negative for every run–layer pair, but magnitude is location dependent. In the corrected cross-family set, weighted layer-mean standard deviation is 0.01746 and the layer range crosses zero. Layer 7 has only three strict matches and is marked low support. Output-geometry adjustment does not reduce this heterogeneity.

H SIGN-REVERSAL ANALYSIS

Twenty-four of 103 strict matched cells have $D_S^{\text{noise}} - D_S^{\text{block}} > 0$, despite the negative aggregate. A frozen geometry classifier reaches leave-one-run-out AUC 0.664 and does not pass the informativeness gate. These reversals are retained and prevent a uniform cellwise family claim.

I OUTPUT-GEOMETRY DETAILS

The preregistered model sequence adds intact margin, anchored boundary displacement, then logit-change norm and absolute alignment. KL and NLL are nearly collinear on the matched set, so the main regression uses a frozen functional-damage summary rather than interpreting their separate coefficients. The full geometry block shrinks the family coefficient by 20.18% ; the geometry-matched subset and reversal analysis provide negative evidence against a complete output-boundary account.

J INTERNAL-DIRECTION NON-IDENTIFIABILITY

The hidden perturbation vectors for the two families are almost orthogonal. Adding observational direction features to the frozen output-geometry baseline improves held-out residual RMSE by 9.63% , below the preregistered 10% gate, and changes reversal AUC by only 0.0026 . The first counterfactual calibration produces $15/103$ jointly supported cells and fails before outcome reveal. The paper therefore makes no statement about the sign or magnitude of a directional causal effect.

K CONTINUOUS-ALPHA FEASIBILITY FAILURE

ARC-009 uses continuous alpha to improve joint support without exposing D_S . Strict PASS-A support rises from 15 to 47 cells and PASS-B reaches 29, but the frozen minimum is not met and layer 2 has $0/15$ PASS-A cells. The causal branch terminates without a reveal; thresholds are not relaxed post hoc.

L REPRODUCIBILITY AND PROVENANCE

Every ARC stores preregistration or protocol documents, raw/processed results, run manifests, commands, integrity hashes, and resource accounting. The Draft 0 provenance table records the exact artifact used for each number. GPU inference is not rerun for manuscript construction. The empirical chain follows the supersession rule $\text{ARC-006} \rightarrow \text{ARC-006R}$; later analyses reproduce the corrected baseline before proceeding.